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PET-derived reactive monomer–dimer additives for efficient toughening of one-component epoxy thermosets

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ABSTRACT

Sustainable and efficient toughening of one-component epoxy systems remains challenging due to inherent trade-offs between fracture resistance, thermal stability, and formulation simplicity. Here, we report a practical drop-in strategy to enhance the toughness of industrial epoxy thermosets using a bis(2-hydroxyethyl) terephthalate monomer–dimer mixture (mBHET) directly derived from glycolyzed polyethylene terephthalate (PET) waste. Without purification or chemical modification, crude mBHET was incorporated into a diglycidyl ether of bisphenol A (DGEBA)/dicyandiamide (DICY) one-component formulation as a reactive diol. At only 5 wt%, mBHET increased tensile toughness and impact strength by approximately 4400% and 91%, respectively, while maintaining glass transition temperature and thermal stability within practical ranges (T_{d5} > 300 °C). Morphological and spectroscopic analyses confirmed covalent network integration without phase separation. Mechanistic studies combining XRD, XPS, and molecular dynamics simulations revealed a stacking-dependent conformational transition of BHET species: flexible dimer structures promote energy dissipation at low loading, whereas excessive aromatic π–π interactions at higher loading induce rigidification and reduced toughness. Importantly, coexistence of monomer and dimer provides synergistic behavior by disrupting excessive stacking while preserving molecular flexibility. This work demonstrates that unrefined PET glycolysis products can function as highly efficient reactive toughening additives, offering a scalable and industrially compatible route toward mechanically robust and more sustainable epoxy materials.

1. Introduction

One-component epoxy systems are widely used as structural adhesives, electronic encapsulants, and composite matrices owing to their excellent storage stability, processing simplicity, and compatibility with large-scale manufacturing [1,2]. In particular, formulations based on diglycidyl ether of bisphenol A (DGEBA) cured with latent hardeners such as dicyandiamide (DICY) represent industrial standards in automotive and electronic applications, where long shelf life and single-step thermal curing are essential [3,4]. However, the highly cross-linked and rigid networks formed during curing render these one-component epoxy systems intrinsically brittle, resulting in low fracture toughness and poor resistance to impact or mechanical overload [5–7]. This brittleness remains a critical limitation for structural applications where long-term mechanical reliability and resistance to crack initiation and propagation are required [7,8].

Extensive efforts have been devoted to improving the toughness of epoxy thermosets, including the incorporation of elastomeric rubbers [9,10], thermoplastic modifiers [11–13], inorganic fillers [14,15], and core–shell particles [16,17]. While these approaches can significantly enhance fracture resistance, they typically rely on phase-separated morphologies formed during curing and are often accompanied by reductions in glass transition temperature or modulus due to matrix plasticization. Flexible diols and soft-segment modifiers have also been explored as reactive toughening agents; however, their effects frequently originate from increased chain mobility and thus involve similar trade-offs in thermal or mechanical rigidity [18,19]. Alternatively, flexible amine-based curing strategies can improve ductility by introducing compliant linkages into the network [20,21], yet their use inevitably converts one-component epoxy formulations into two-component systems, thereby sacrificing storage stability and processing convenience. Consequently, despite substantial progress, most

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reported approaches remain constrained by inherent trade-offs between toughness, thermal stability, and formulation simplicity (Fig. S1a) [9,22]. Achieving substantial toughening while retaining the thermal and structural integrity of one-component epoxy systems thus remains a persistent and practically important challenge.

From an industrial standpoint, an ideal toughening strategy should be directly applicable to existing commercial epoxy formulations without requiring redesign of the resin, curing agent, or processing protocol [9]. However, most previously reported methods involve structural modification of epoxy monomers, alteration of curing chemistry, or introduction of polymeric modifiers that are not readily compatible with established industrial platforms [23]. The development of a simple reactive additive that can be incorporated into widely used one-component epoxy systems while maintaining formulation architecture would therefore represent a highly attractive and immediately implementable solution.

Alongside these materials challenges, growing attention has been directed toward sustainable polymer additives derived from recycled plastics. Bis(2-hydroxyethyl) terephthalate (BHET), obtained via glycolysis of polyethylene terephthalate (PET), has emerged as a promising platform molecule owing to its aromatic backbone and bifunctional hydroxyl groups [24,25]. Although BHET has primarily been utilized as a precursor for repolymerization [26,27] or for the synthesis of polyesters [28] and polyurethanes [29,30], its direct use as a reactive toughening additive for structural epoxy systems has not been systematically investigated. In particular, crude BHET products generated from PET glycolysis typically consist of a mixture of oligomeric species, in which monomer and dimer are the dominant components whose roles in epoxy network formation and mechanical performance remain largely unexplored.

Herein, we present a practical and sustainable toughening strategy for one-component epoxy systems by directly utilizing a BHET monomer–dimer mixture powder (mBHET) derived from PET bottle glycolysis as a reactive diol additive. Unlike conventional PET upcycling approaches that require purification to isolate monomeric species or chemical transformation of monomer, the mBHET can be introduced directly into an industrially relevant DGEBA/DICY epoxy formulation without altering the established curing system (Fig. 1a, S1b). This drop-in strategy enables substantial enhancement of fracture resistance and impact tolerance at low additive loadings while largely preserving the thermal stability and processing advantages of one-component epoxy systems (Fig. 1b). Comparative studies of BHET monomer, dimer, and their mixtures, together with molecular dynamics (MD) simulations, reveal that the monomer–dimer coexistence plays a key role in promoting energy dissipation within the epoxy network. These findings demonstrate that a simple additive derived from PET bottle waste can be directly integrated into commercial epoxy formulations to achieve effective toughening, providing a sustainable and industrially practical route toward mechanically robust epoxy thermosets while mitigating the conventional trade-offs associated with epoxy toughening strategies. While further process considerations, such as solvent recovery and optimization of glycolysis conditions, remain pivotal for large-scale implementation, the present approach demonstrates the feasibility of directly employing unrefined glycolysis products as functional additives.

2. Results & discussion

2.1. Design strategy

In this study, the formulation strategy was guided by two key considerations: industrial relevance and systematic evaluation of additive efficiency. A representative one-component epoxy system composed of DGEBA as the resin, DICY as the latent hardener, and U-52 M as the accelerator was selected, as all components are commercially available and widely employed in structural adhesive and electronic encapsulation applications. In particular, DICY enables storage-stable one-component epoxy formulations that are essential for large-scale manufacturing processes such as automotive assembly and electronic packaging [3,4]. However, the highly crosslinked networks formed in DICY-cured systems typically result in intrinsically brittle mechanical behavior when no additional modifiers are used, and a wide range of reactive and non-reactive toughening approaches have therefore been explored to improve fracture resistance [31–35].

Despite these efforts, many previously reported strategies involve either non-reactive modifiers that rely on phase-separated morphologies or curing-system modifications that alter formulation simplicity, often accompanied by trade-offs in thermal or mechanical integrity [9,36]. To address these limitations while preserving the advantages of one-component epoxy systems, we adopted a drop-in additive strategy in which a reactive diol derived from PET glycolysis was directly incorporated into an established industrial formulation without altering the curing chemistry.

Accordingly, mBHET was introduced as a reactive toughening additive into the DICY-cured epoxy system, and its concentration was systematically varied from 5 to 30 wt% (denoted as EBM-5 to EBM-30) (Table S1). This formulation provides a practical platform for evaluating the effectiveness of a PET-derived reactive additive under industrially

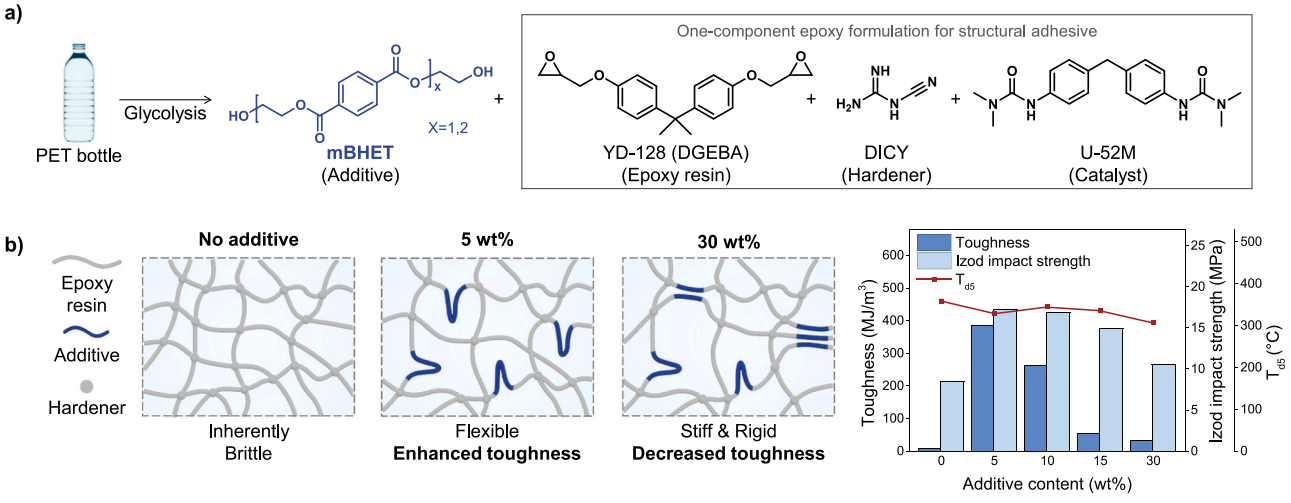


Fig. 1. (a) Schematic illustration of the glycolysis reaction used to depolymerize PET bottles in this study and the epoxy formulation employed. (b) Toughening effect of mBHET additives on epoxy resin as a function of additive content.

relevant one-component curing conditions, enabling direct assessment of its ability to enhance fracture resistance while maintaining formulation simplicity and structural integrity.

2.2. Glycolysis of PET to obtain mBHET

The glycolysis process used to depolymerize PET into BHET species is illustrated in Fig. S2a. In conventional PET upcycling approaches, BHET recovered from glycolysis is typically purified to isolate the monomer prior to reuse as a building block for polymer synthesis [37]. However, such purification often reduces overall yield and increases processing cost, limiting the practicality of direct recycling strategies [37]. In the present study, this purification step to isolate the monomer was intentionally avoided, and the as-obtained glycolysis product was directly utilized as a functional additive.

Nuclear magnetic resonance (NMR) analysis confirmed the successful depolymerization of PET into mBHET (Fig. S3, S4 and S5). Characteristic signals corresponding to the monomer and dimer appeared at 8.1 and 4.6 ppm, respectively, and integration of these peaks revealed a monomer-to-dimer ratio of approximately 1:0.4 (Fig. S6) [38]. No distinct signals attributable to higher oligomer species were observed, suggesting that they are present only in trace amounts. This observation is further supported by differential scanning calorimetry (DSC) analysis, in which no additional transitions associated with higher oligomers were detected (Fig. S8). As this monomer-to-dimer composition was consistently obtained under our glycolysis conditions, the epoxy formulations were subsequently prepared using the resulting mBHET without further alteration. Consequently, the mechanical behavior of the system was interpreted primarily based on the contributions of monomer and dimer species.

The compositional heterogeneity is an intrinsic feature of glycolysis-derived BHET and was hypothesized to influence network formation and mechanical performance when incorporated into epoxy systems. To elucidate the individual contributions of each component and clarify the role of monomer–dimer synergy, BHET monomer and dimer were separately used to formulate corresponding epoxy systems. These comparative formulations enabled systematic evaluation of the toughening behavior arising from each component and from the crude mBHET mixture (vide infra), providing insight into the structure–property relationships governing the performance of PET-derived reactive additives in one-component epoxy networks.

2.3. Curing behavior and integration of mBHET into the epoxy network

For a toughening additive to be practically useful in structural epoxy systems, it must be chemically integrated into the crosslinked network rather than acting as a simple plasticizer. Our previous work demonstrated that diol-type additives can be covalently incorporated into epoxy matrices during curing, as confirmed by DSC, Fourier-transform infrared spectroscopy (FT-IR), and swelling analyses [18]. Building upon this methodology, we investigated whether the PET-derived mBHET additive actively participated in network formation under the present one-component curing conditions.

To determine whether mBHET was chemically incorporated into the epoxy network, DSC measurements were performed with particular focus on the second heating cycle after curing (Fig. S7). During the initial heating cycle, two endothermic transitions were observed: a sharp melting peak near 110 °C corresponding to the BHET monomer and a broader transition with an onset near 170 °C associated with the BHET dimer (Fig. S8). Upon subsequent cooling and reheating, only a single melting peak attributable to recrystallized dimer was detected, as the monomer is less prone to recrystallization under the rapid thermal conditions of the second cycle [39,40].

If mBHET had remained unreacted after curing, characteristic melting transitions of the monomer or dimer would have reappeared during the second heating cycle. However, as shown in Fig. S7b, no melting endotherm was observed upon reheating, indicating that the mBHET species had fully reacted and were chemically integrated into the epoxy network. This conclusion is further supported by the absence of residual crystalline domains and the transparent, homogeneous appearance of the fully cured specimens (Fig. S2b). Together, these results confirm that mBHET functions as a reactive network-integrating additive rather than a phase-separating plasticizer, enabling molecular-level modification of the epoxy structure while preserving the integrity of the one-component curing system.

The curing behavior of the EBM samples was further examined by DSC to elucidate how the PET-derived additive influences network formation (Fig. S9, Table S2). Epoxy curing proceeds through an exothermic reaction, and the evolution of heat flow provides insight into reaction pathways and kinetics [41]. The neat epoxy formulation exhibited a single sharp exothermic peak corresponding to the curing reaction between DGEBA and DICY. In contrast, EBM samples displayed additional thermal features. An endothermic transition near 90 °C was first observed and attributed to melting of residual mBHET, followed by a broad exothermic peak and a distinct high-temperature shoulder whose intensity increased with increasing mBHET content. The emergence of this shoulder peak indicates that incorporation of mBHET modifies the curing pathway and alters the relative contributions of competing reactions within the epoxy system.

Two possible origins were considered for this shoulder peak: (i) secondary reactions between hydroxyl groups of mBHET and residual epoxy groups after primary curing, or (ii) homopolymerization (self-condensation) of unreacted epoxy groups at elevated temperatures. To distinguish between these mechanisms, model curing experiments were conducted using systems containing either epoxy resin alone or epoxy resin with mBHET in the presence of catalyst (Fig. S10). The exothermic signal corresponding to the epoxy–mBHET reaction was observed at approximately 160 °C, whereas epoxy self-condensation occurred at higher temperature (~180 °C). Because the shoulder peak in the EBM formulations coincided with the latter temperature range, it was assigned primarily to homopolymerization of residual epoxy groups.

The main curing peak centered near 162 °C is therefore attributed to concurrent reactions of epoxy groups with DICY and mBHET, while the higher-temperature shoulder arises from subsequent self-condensation of remaining epoxy functionalities. The presence of aromatic BHET units introduces steric constraints within the reacting network, which may partially hinder access of DICY to epoxy groups during the early stages of curing [42]. This steric effect can delay complete consumption of epoxy groups and promote secondary homopolymerization at elevated temperatures. Consequently, incorporation of mBHET not only enables its covalent integration into the network but also subtly modifies curing kinetics and reaction pathways, leading to a network architecture distinct from that of the neat epoxy. Such molecular-level changes in network formation are expected to influence energy dissipation behavior and contribute to the observed toughening of the one-component epoxy system.

The chemical incorporation of mBHET into the epoxy network was further verified by swelling and spectroscopic analyses. For swelling tests, fully cured samples were immersed in acetone—a good solvent for mBHET—for three days, followed by drying and weighing before and after immersion (Fig. S11). No measurable weight loss was detected for any EBM sample, indicating that mBHET was not extractable and had been covalently integrated into the crosslinked epoxy network rather than remaining as a physically dispersed component.

FT-IR spectroscopy provided additional molecular-level evidence for incorporation of mBHET into the epoxy matrix (Fig. S12). The characteristic oxirane ring absorption at 910 cm^{-1}, associated with epoxy groups, disappears upon curing due to ring-opening reactions [43,44]. Both neat epoxy and EBM samples exhibited near-complete disappearance of this band, confirming effective curing of the epoxy system.

The FT-IR spectrum of the mBHET powder displayed a strong ester C═O stretching vibration near 1715 cm^{-1} and a broad O—H stretching

band around 3430 cm^{-1}, consistent with the presence of aromatic ester and terminal hydroxyl functionalities derived from PET glycolysis [45]. In the cured EBM samples, the ester C═O stretching peak remained clearly visible, indicating that the aromatic ester structure of mBHET was preserved during curing and incorporated into the network without chemical degradation. All cured samples exhibited broad O—H stretching bands, which can be attributed to hydroxyl groups generated during epoxy ring opening as well as residual hydroxyl end groups from incorporated mBHET.

Taken together, the swelling and FT-IR results provide complementary evidence that mBHET functions as a reactive, network-integrating additive rather than a phase-separating plasticizer. The preservation of aromatic ester functionality alongside complete epoxy conversion suggests that the PET-derived additive is covalently incorporated while maintaining its structural integrity, thereby enabling molecular-level modification of the epoxy network without compromising curing efficiency.

2.4. Effect of mBHET content on toughness

To evaluate the toughening efficiency of the PET-derived additive, the mechanical properties of epoxy formulations containing varying amounts of mBHET were investigated. Based on our previous study demonstrating that diol-type additives can enhance toughness through network modification [18], we hypothesized that incorporation of aromatic BHET species would similarly promote energy dissipation while maintaining structural integrity.

Stress–strain curves of the neat epoxy and EBM samples were obtained using a universal testing machine (UTM) (Fig. 2a), and tensile toughness was calculated from the area under each curve (Table 1) [46]. The neat epoxy exhibited typical brittle fracture behavior characterized by limited elongation and low toughness. The tensile strength of the neat epoxy is at the lower end of reported values in DGEBA/DICY systems (Table S4), which is known to vary widely depending on formulation and curing conditions [47]. Upon incorporation of mBHET, both tensile stress and strain at break increased, indicating enhanced resistance to crack initiation and propagation. At 5 wt% mBHET loading, the tensile strength and strain increased from 13.7 MPa and 1.20% (neat epoxy) to 70.4 MPa and 9.84%, respectively, resulting in a substantial rise in toughness from 8.58 to 385.6 $MJ{\cdot}m^{-3}$. Beyond this point, a gradual decline in mechanical performance was observed at higher contents. While our previous work required ~30 wt% diol additive to reach maximum toughness [18], the present formulation attained a

Table 1
Summary of key mechanical properties for each formulation.

Sample name	Tensile strength (MPa)	Tensile strain (%)	Toughness (MJ/m^3)	Izod impact strength (J/m)
Neat epoxy	13.70 ± 1.90	1.20	8.58	8.53 ± 1.65
EBM-5	70.41 ± 3.20	9.84	385.58	16.30 ± 1.31
EBM-10	58.57 ± 6.09	8.52	262.70	15.05 ± 0.49
EBM-15	25.94 ± 1.96	4.22	55.17	14.88 ± 2.21
EBM-30	22.91 ± 3.36	2.50	32.86	10.52 ± 2.16

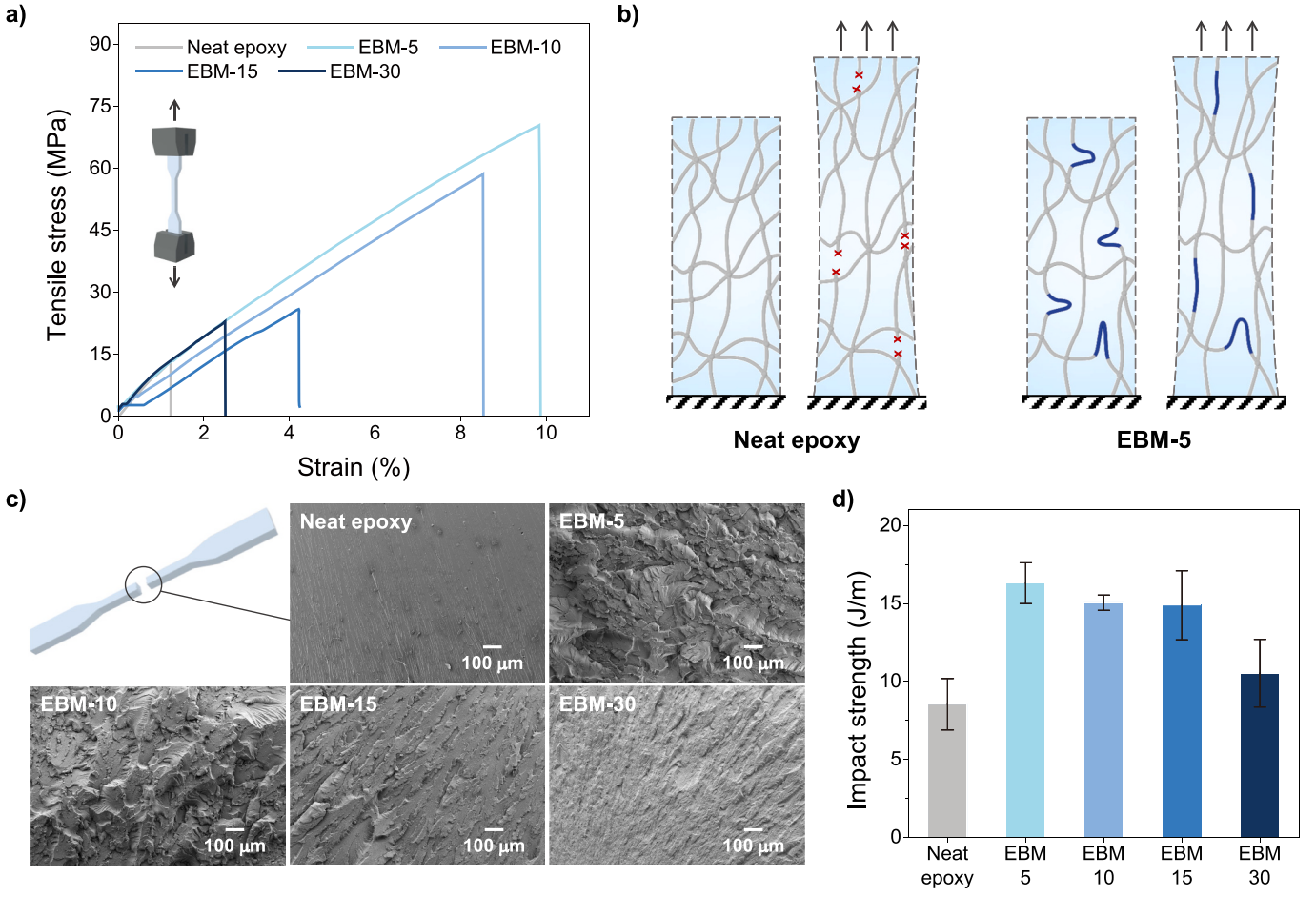


Fig. 2. (a) Tensile stress–strain curves of EBM samples. The area under the curve (toughness) increases with mBHET content up to an optimal level. (b) Schematic representation of chain deformation behavior in Neat epoxy and EBM-5 during tensile testing. (c) SEM images of tensile fracture surfaces, showing morphological changes induced by mBHET incorporation. (d) Izod impact strength as a function of mBHET loading.

comparable toughness value of 385.6 MJ·m^{-3} with only 5 wt% mBHET. This corresponds to an approximately 4400% increase relative to the neat epoxy and exceeds improvements typically reported across a broad range of epoxy toughening systems—including those based on bio-based additives, inorganic fillers, or conventional reactive modifiers (Fig. S13, Table S3). For a more direct comparison within the same matrix system, representative toughness improvements reported for DGEBA/DICY-based epoxy systems are separately summarized in Table S4, where the present system also demonstrates competitive performance. Such pronounced enhancement at low loading suggests that the intrinsic monomer–dimer composition of mBHET provides a particularly efficient toughening pathway.

Under tensile deformation, crack initiation and propagation in brittle epoxy networks are governed by localized bond scission and limited chain mobility [48]. Introduction of reactive diol additives can modify the network architecture by introducing flexible segments and additional energy dissipation pathways, thereby promoting more uniform plastic deformation and delaying catastrophic failure. In the present system, the combination of aromatic rigidity and flexible ether linkages within mBHET is expected to facilitate localized stress redistribution while maintaining network cohesion. In particular, BHET dimers, with their bent and extended structures, may act as molecular-level energy dissipaters that enable stress transfer and suppress rapid crack propagation (Fig. 2b).

Importantly, excessive additive loading can adversely affect network homogeneity by introducing steric congestion or localized heterogeneity, which may restrict chain mobility or promote microphase separation [49]. A similar trend was observed here: when mBHET content exceeded the optimal level, toughness gradually decreased, indicating that efficient toughening arises from a balanced incorporation of mBHET rather than simple concentration-dependent softening. These results highlight that low-level incorporation of PET-derived aromatic diols can substantially improve fracture resistance while avoiding the plasticization and phase-separation issues commonly associated with conventional toughening strategies.

2.5. Fracture morphology and impact resistance

To further elucidate the toughening mechanism, fracture surfaces of tensile-tested specimens were examined by scanning electron microscopy (SEM) (Fig. 2c). The neat epoxy exhibited a smooth and featureless fracture surface characteristic of brittle failure, indicating rapid crack propagation with minimal energy dissipation. In contrast, all mBHET-containing samples displayed markedly rough and irregular fracture morphologies. The EBM-5 formulation, which exhibited the highest tensile toughness, showed the most pronounced surface roughness and tortuosity.

Such roughened fracture surfaces are indicative of activated energy-dissipation mechanisms during crack propagation, including crack deflection, localized plastic deformation, and increased fracture path tortuosity [49]. These features suggest that incorporation of mBHET promotes more distributed stress relaxation within the network, thereby delaying catastrophic fracture. Notably, no cavitation or discrete phase domains were observed in any EBM sample, supporting that mBHET is covalently integrated into the epoxy network rather than acting as a phase-separating rubbery modifier [42]. This molecular-level integration enables localized deformation without compromising network continuity, resulting in enhanced fracture resistance.

Izod impact testing revealed a similar trend (Fig. 2d). Impact strength improved with mBHET incorporation, reaching 16.3 J·m^{-1} at 5 wt% loading, which corresponds to a ~ 92% increase relative to the neat epoxy (8.53 J·m^{-1}). This value exceeds that obtained in our previous diol-based system even at higher additive loadings, highlighting the high toughening efficiency of mBHET at low concentration. Beyond 5 wt % loading, further addition of mBHET led to a gradual decrease in impact strength.

Under impact loading, efficient energy dissipation requires a network capable of redistributing localized stress and suppressing rapid crack propagation. Incorporation of mBHET introduces molecular segments that facilitate localized plastic deformation and stress redistribution while maintaining network connectivity. However, excessive additive incorporation can perturb network homogeneity or introduce steric constraints that hinder efficient stress transfer, resulting in reduced impact resistance [9]. The observed optimum at low mBHET content therefore reflects a balance between enhanced energy dissipation and preservation of network integrity.

Taken together with tensile results, these findings demonstrate that PET-derived mBHET provides highly efficient toughening at low loading without relying on phase-separated morphologies typically associated with rubber or thermoplastic modifiers. The ability to achieve substantial improvements in fracture and impact resistance while maintaining a homogeneous network structure underscores the effectiveness of this reactive, drop-in toughening strategy for one-component epoxy systems.

2.6. Effect of residual catalyst and trace oligomers

Residual catalyst and trace oligomeric species in the mBHET obtained from PET glycolysis may pose concerns regarding their influence on the epoxy system. Inductively coupled plasma mass spectrometry (ICP-MS) analysis was performed to quantify residual catalyst species, confirming that Zn remains at approximately 0.32 wt%, originating from the zinc-based catalyst used during glycolysis. Such residual metal species could, in principle, affect curing behavior by acting as Lewis acid catalysts, thereby influencing epoxy ring-opening reactions [50]. In addition, trace amounts of higher oligomers may also be present, although their contribution is expected to be minimal based on NMR and DSC analyses, which indicate that monomer and dimer are the dominant components. To evaluate whether these residual components significantly influence the mechanical properties, control experiments were conducted using synthetically blended BHET monomer–dimer mixtures with a composition identical to that of mBHET (monomer:dimer ≈ 1:0.4). The resulting epoxy systems exhibited tensile strength and Izod impact strength comparable to those of the mBHET-based system (Fig. S14), indicating that the observed mechanical behavior is primarily governed by the monomer–dimer composition rather than by residual catalyst or trace oligomeric species. Therefore, the effect of residual impurities on the overall properties is considered negligible in the present system.

2.7. Origin of toughness reduction at high mBHET loadings: suppression of flexibility by aromatic interactions

Although incorporation of mBHET significantly improved toughness at low loading, a gradual decline in mechanical performance was observed at higher concentrations. To elucidate the origin of this behavior, we examined whether the reduction in toughness arose from phase separation or from molecular-level interactions between aromatic moieties of mBHET within the epoxy network.

We first investigated the possibility of phase separation. Atomic force microscopy (AFM) images of neat epoxy and mBHET-containing samples with varying loadings are shown in Fig. 3a and Fig. S15. All samples exhibited homogeneous surface morphology without discernible phase-separated domains or microstructural heterogeneity. Consistently, small-angle X-ray scattering (SAXS) analysis revealed no characteristic scattering peaks or shoulders associated with periodic domain formation (Fig. 3b) [51]. These results collectively indicate that the decrease in toughness at high additive loading does not originate from macroscale or microscale phase separation.

We next examined whether intermolecular interactions among aromatic units of mBHET could influence network mechanics. X-ray diffraction (XRD) patterns of the PET precursor displayed a diffraction peak near 27° (2θ), corresponding to a π–π stacking distance of

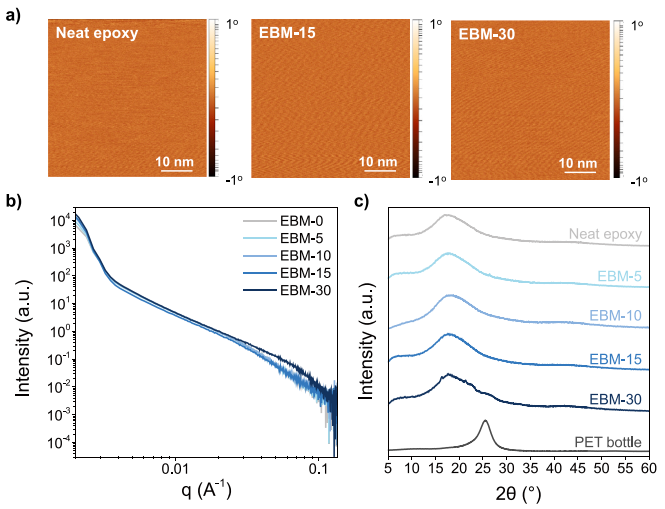


Fig. 3. (a) AFM phase images of Neat epoxy, EBM-15, and EBM-30 samples. (b) SAXS profiles of Neat epoxy and EBM formulations. (c) XRD patterns of Neat epoxy, EBM samples, and the PET bottle used in the glycolysis reaction.

approximately 3.9–4.1 Å [52,53]. Interestingly, EBM samples with high mBHET loading exhibited a similar shoulder peak in this region (Fig. 3c), suggesting the emergence of partial aromatic stacking interactions within the cured network. Such features were absent at low mBHET concentrations, indicating that these interactions become significant only beyond a certain loading threshold.

X-ray photoelectron spectroscopy (XPS) provided complementary evidence. The C 1 s spectrum of the high-loading sample (mBHET-30) displayed an intensified satellite peak at ~291 eV corresponding to π–π* shake-up transitions (Fig. S16, Table S5) [54], confirming enhanced aromatic interactions at elevated mBHET content. These observations suggest that close packing of aromatic terephthalate units promotes intermolecular π–π interactions within the epoxy network at high additive concentrations.

Such interactions can restrict conformational mobility of mBHET segments and increase local network rigidity [55]. As a result, the hinge-like flexibility that facilitates stress redistribution at low additive loading becomes progressively constrained at higher concentrations. The additive therefore transitions from a molecular energy-dissipation element to a rigidifying component that limits localized plastic deformation, leading to reduced toughness.

Taken together, these findings indicate that efficient toughening arises from an appropriate incorporation of mBHET in which molecular flexibility and network integration are maintained. At low loading, mBHET introduces effective energy-dissipation pathways without disrupting network homogeneity. At excessive loading, however, increased aromatic interactions impose conformational constraints that elevate stiffness and diminish fracture resistance. This molecular-level insight highlights the importance of additive–additive interactions in governing the toughening efficiency of aromatic reactive modifiers in epoxy systems.

2.8. Molecular dynamics insights into stacking-dependent toughening behavior

To gain molecular-level insight into how BHET architecture influences network mechanics, MD simulations were performed to evaluate the effects of molecular stacking on conformation and energy dissipation behavior. Amorphous three-dimensional BHET assemblies with varying degrees of intermolecular stacking were constructed and equilibrated under canonical ensemble conditions to obtain representative conformations (Fig. S17).

Analysis of chain conformations revealed that BHET dimers adopt significantly more folded and coiled structures than monomeric species due to the torsional flexibility of the O–C–C–O dihedral linkage connecting aromatic units (Fig. 4a, b). These folded conformations act as molecular “hinges” that can extend and reconFig. under mechanical deformation, thereby enabling efficient stress redistribution and energy dissipation within the epoxy network [9,56]. Such flexibility provides a molecular basis for the superior toughening efficiency observed at low mBHET loadings, where dispersed dimeric species can function as nanoscale energy-dissipating elements.

Interestingly, as the degree of intermolecular stacking increased, the conformational landscape of BHET dimers shifted from folded and flexible to more extended and rigid configurations. This transition arises from π–π interactions between neighboring aromatic rings, which restrict torsional mobility and increase the effective bending stiffness of the assemblies. The resulting rod-like conformations limit conformational adaptability and reduce the ability of the additive to dissipate mechanical energy through structural rearrangement [57]. Instead, applied stress is increasingly localized within the molecular backbone, leading to elevated internal strain and reduced macroscopic toughness. This stacking-induced stiffening provides a molecular-level explanation for the experimentally observed decline in toughness at high mBHET loadings.

To further evaluate how these conformational changes influence network-level mechanical response, highly crosslinked epoxy models containing BHET additives with varying stacking states were constructed (Fig. 4c, S18). Strain energy evolution during simulated deformation revealed that incorporation of BHET generally lowered strain energy density relative to the hardener-dominated network, confirming its role as an energy-dissipation component (Fig. 4d). However, the magnitude and distribution of strain energy were strongly dependent on molecular architecture. Non-stacked, flexible BHET dimers delayed strain accumulation and distributed mechanical load more uniformly across the network, whereas stacked assemblies concentrated strain within rigid segments, accelerating local damage formation.

Atomic-level analysis of strain energy distribution within the BHET-dimer backbone further highlighted the role of torsional flexibility (Fig. 4e). In stacked configurations, strain energy density was significantly higher along the O–C–C–O dihedral linkage, indicating restricted rotational freedom and amplified molecular stress under deformation. In contrast, non-stacked configurations retained greater conformational freedom, enabling more homogeneous stress dispersion and reduced structural degradation.

These simulation results collectively demonstrate that the toughening efficiency of PET-derived mBHET additives is governed not only by their chemical incorporation into the network but also by their intermolecular organization. At low loading, dispersed and conformationally flexible BHET species enhance energy dissipation and delay crack propagation. At high loading, increased aromatic stacking constrains molecular mobility and promotes stiffness, thereby diminishing toughness. This stacking-dependent flexibility transition provides a consistent molecular-level explanation for the experimentally observed optimum in mechanical performance at low additive concentrations.

2.9. Synergistic toughening mechanism of monomer–dimer mixture additives

To further clarify why the mBHET provides superior toughening compared to either component alone, we investigated the thermal and structural stability of BHET species as a function of molecular length using MD simulations and complementary experiments.

Crystalline assemblies composed exclusively of BHET monomers or dimers were constructed and equilibrated under isothermal–isobaric conditions at 300 K and 1 atm (Fig. 5a). A linear temperature ramp (10 K ns^{-1}) up to 450 K—exceeding the epoxy curing temperature—was then applied while monitoring density changes (Fig. 5b). Both systems exhibited gradual thermal expansion; however, the monomer-based

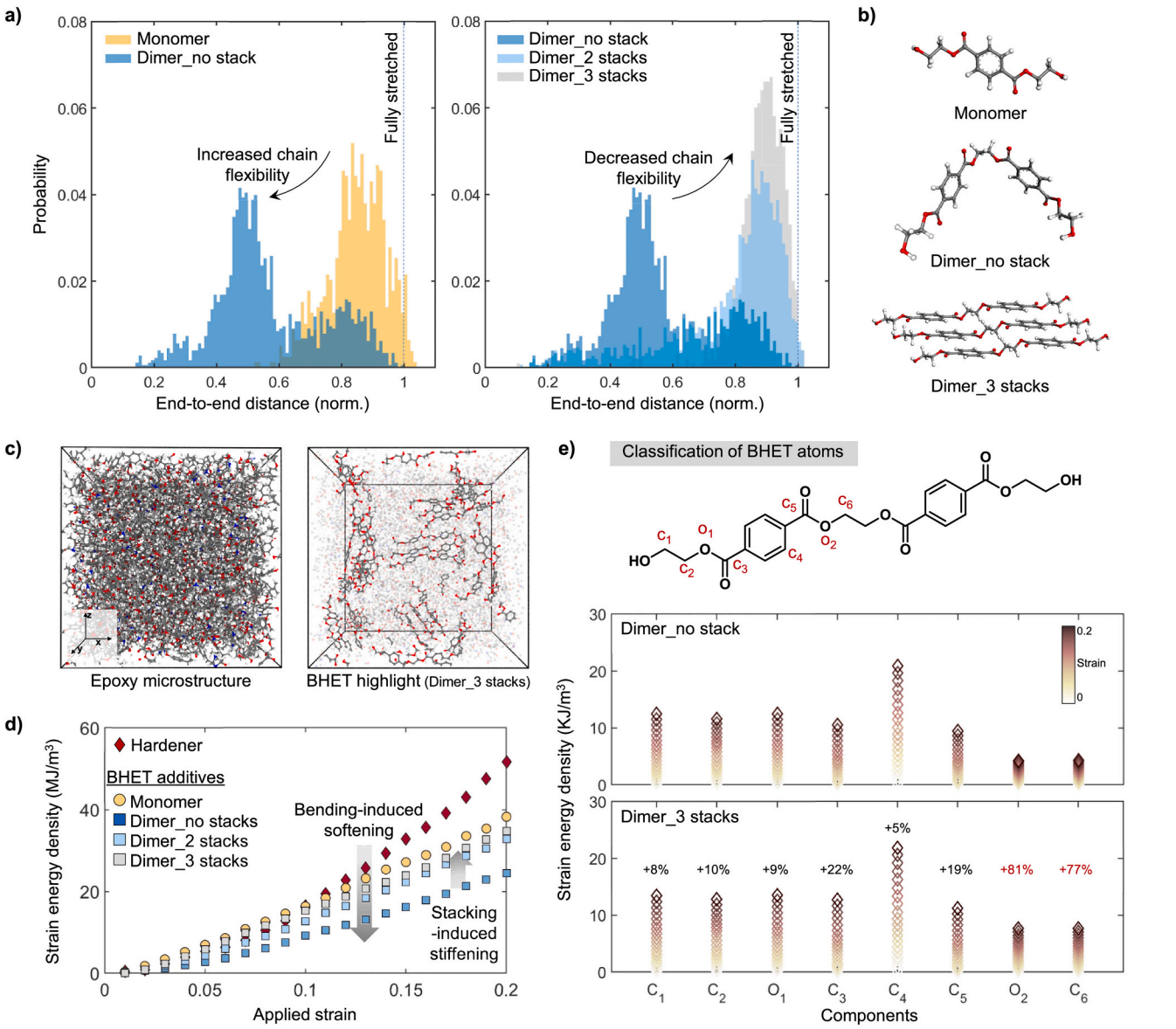


Fig. 4. Estimation of epoxy network morphology according to the content of mBHET and conceptual illustration of the proposed mechanism for toughness reduction with increasing mBHET content. (a) Variations in the end-to-end distance of BHET molecules as a function of chain length and stacking levels. (b) Molecular configurations of BHET monomer and dimer in non-stacked and stacked forms. (c) All-atomistic configuration of BHET-reinforced epoxy microstructure. (d) Mechanical strain energy density evolved in BHET chains upon uniaxial tension. (e) Subdivision of strain energy density for BHET backbone constituents. The quantitative ratio in strain energy density between no stack and 3 stacks models is also presented.

model showed an abrupt density drop near 380 K, indicating rapid disruption of intermolecular interactions at elevated temperature. In contrast, dimer-based assemblies maintained continuous density profiles across the temperature range, reflecting stronger intermolecular cohesion and enhanced thermal robustness (Fig. S19). These results suggest that while BHET monomers readily dissociate at high temperature, dimers retain partial stacking owing to their dual aromatic rings and extended π systems.

Interestingly, mixed monomer–dimer assemblies exhibited earlier dissociation than pure dimer crystals, with density changes occurring near 398 K. This behavior indicates that monomer incorporation introduces packing frustration and disrupts ordered stacking among dimer species. The terminal hydroxyl groups of monomers promote local thermal fluctuations and act as structural perturbation sites, thereby preventing excessive aggregation of aromatic dimer units. Such disruption of stacking is expected to preserve conformational flexibility within the additive ensemble.

Experimental impact strength measurements support this interpretation (Fig. 5c). Epoxy samples containing only monomer or only dimer exhibited improved toughness relative to neat epoxy, consistent with the role of diol additives in modifying network architecture. The dimer provided higher impact strength than the monomer, reflecting its hinged molecular structure that facilitates energy dissipation. However, the performance of the dimer-only system remained inferior to that of the monomer–dimer mixture. This reduced efficiency likely arises from partial stacking of dimer molecules, even at moderate concentrations, which restricts their conformational freedom and limits energy dissipation. At higher loadings, increased stacking further stiffens the network and diminishes toughness. It should be noted that dimer at 30 wt% could not be incorporated into epoxy samples due to its limited processability, reinforcing the contribution of the monomer to improving processability.

In contrast, the presence of monomer units disrupts excessive dimer–dimer interactions, maintaining molecular mobility while

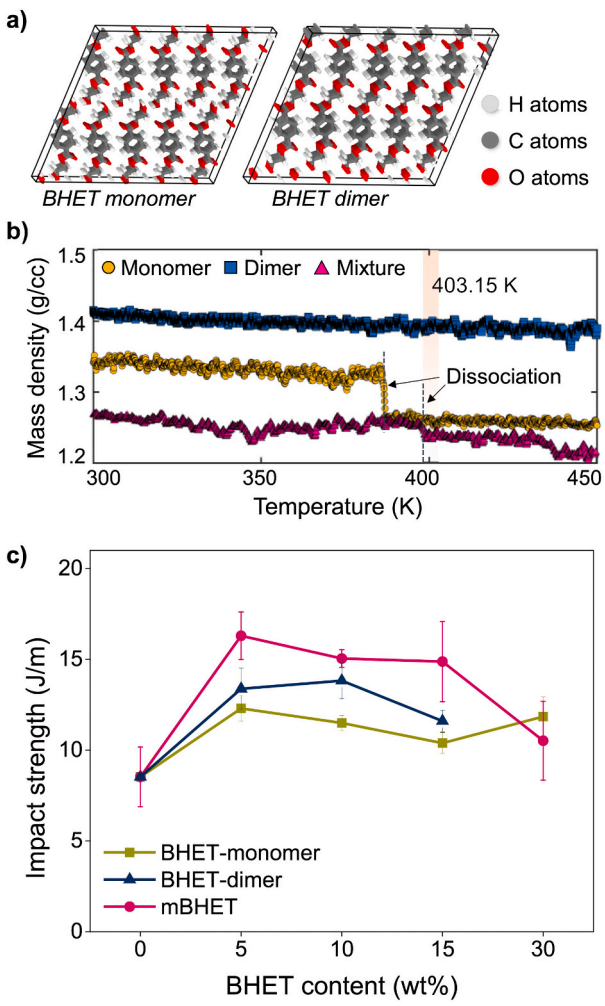


Fig. 5. (a) Crystal structures of BHET-monomer and dimer prepared in molecular dynamics simulation. (b) Structural stability analysis of BHET crystals under a gradual temperature rise. (c) Izod impact strength of epoxy samples as a function of additive content, comparing formulations containing BHET monomer, dimer, and a monomer–dimer mixture.

improving dispersion and processability. The monomer–dimer mixture therefore combines these complementary effects: dimer units provide effective energy-dissipation pathways through their flexible hinge-like structures, while monomer units act as molecular spacers that suppress excessive aromatic stacking and preserve conformational freedom. This synergistic interplay results in superior mechanical performance compared with either component alone.

Taken together, these findings reveal that the toughening efficiency of glycolysis-derived mBHET originates from the coexistence of monomer and dimer species. Without additional purification, mBHET intrinsically provides a molecular architecture in which monomer-induced disruption suppresses excessive stacking-induced stiffening of dimers. Notably, the degree of improvement depended on additive type and content (Fig. 5c) and the Izod impact strength varied with different monomer–dimer ratios for EBM-5 formulation (Fig. S20). These results indicate that the monomer-to-dimer ratio is another important factor influencing toughening behavior; however, identifying the optimal composition was beyond the scope of the present study and remains a subject for future investigation.

2.10. Mechanical and thermal performance of mBHET-modified epoxy systems

The overall mechanical and thermal performance of the cured epoxy systems was evaluated to assess the practical impact of incorporating mBHET as a reactive toughening additive. Dynamic mechanical analysis (DMA) revealed the temperature dependence of storage modulus (E') and tan δ for neat and modified systems (Fig. 6a). The glass transition temperature (T_g) was determined from the tan δ peak ($T_{g,DMA}$) and further verified by DSC measurements ($T_{g,DSC}$) (Fig. 6b) [58].

With increasing mBHET content, T_g gradually decreased, consistent with a reduction in effective crosslinking density and the accompanying increase in segmental mobility [59,60]. In DICY-cured epoxy systems, the hardener can theoretically form multiple crosslinks, resulting in a densely crosslinked and rigid network [61]. Incorporation of mBHET introduces additional hydroxyl functionalities that compete with DICY for epoxy groups and introduces longer molecular segments within the network, leading to a controlled reduction in crosslink density and enhanced chain mobility.

To quantify this structural change, the effective crosslinking density (ρ) was estimated from the storage modulus in the rubbery plateau region (E'_e) using the rubber elasticity relationship:

$$\rho = \frac{E'_e}{3RT}$$

where R is the gas constant, and T is the absolute temperature in the rubbery region [43,60].

As summarized in Table 2, the calculated crosslinking density decreased progressively with increasing mBHET content, consistent with the T_g trend. Importantly, however, the thermal penalty at the optimum toughening condition was small: at EBM-5, where toughness increased dramatically, the T_g reduction was only modest. This behavior contrasts with many conventional toughening strategies—particularly those based on flexible soft segments or plasticization—where significant toughness gains are typically accompanied by large T_g drops and substantial losses in structural integrity [9]. Instead, the present system achieves pronounced toughening at low loading while largely preserving the thermal properties required for practical one-component epoxy applications, indicating that mBHET mitigates (rather than exacerbates) the typical toughness–T_g trade-off.

At higher mBHET loadings, the calculated crosslink density of EBM-30 deviated from the decreasing trend and appeared anomalously high. This discrepancy likely arises from increased physical associations such as π–π interactions between aromatic moieties, which elevate the rubbery plateau modulus and result in overestimation of crosslink density despite a chemically less crosslinked network [62]. These findings further support that, at higher mBHET loadings, aromatic stacking contributes to rigidification and reduced toughness, consistent with the structural and spectroscopic evidence discussed above.

Thermogravimetric analysis (TGA) confirmed that incorporation of mBHET leads to only moderate changes in thermal stability (Fig. 6c). As expected, decreasing crosslink density resulted in a gradual reduction in T_{d5} with increasing additive content [63]. Notably, at 5 wt% mBHET the T_{d5} decrease relative to neat epoxy was minimal (9%), substantially smaller than reductions previously observed for systems employing flexible aliphatic diol additives at their optimal toughening conditions (T_{d5} 23% decrease at 30 wt%) [18]. The improved thermal retention is attributed to the aromatic structure of mBHET, whose high bond dissociation energies help preserve network stability despite reduced crosslink density [64]. Meanwhile, all formulations maintained T_{d5} values above 300 °C, confirming sufficient thermal robustness for structural and adhesive applications.

Lap shear strength tests were conducted to evaluate the applicability of mBHET as an additive for structural epoxy adhesive (Fig. 6d). A gradual decrease in shear strength with increasing mBHET content was

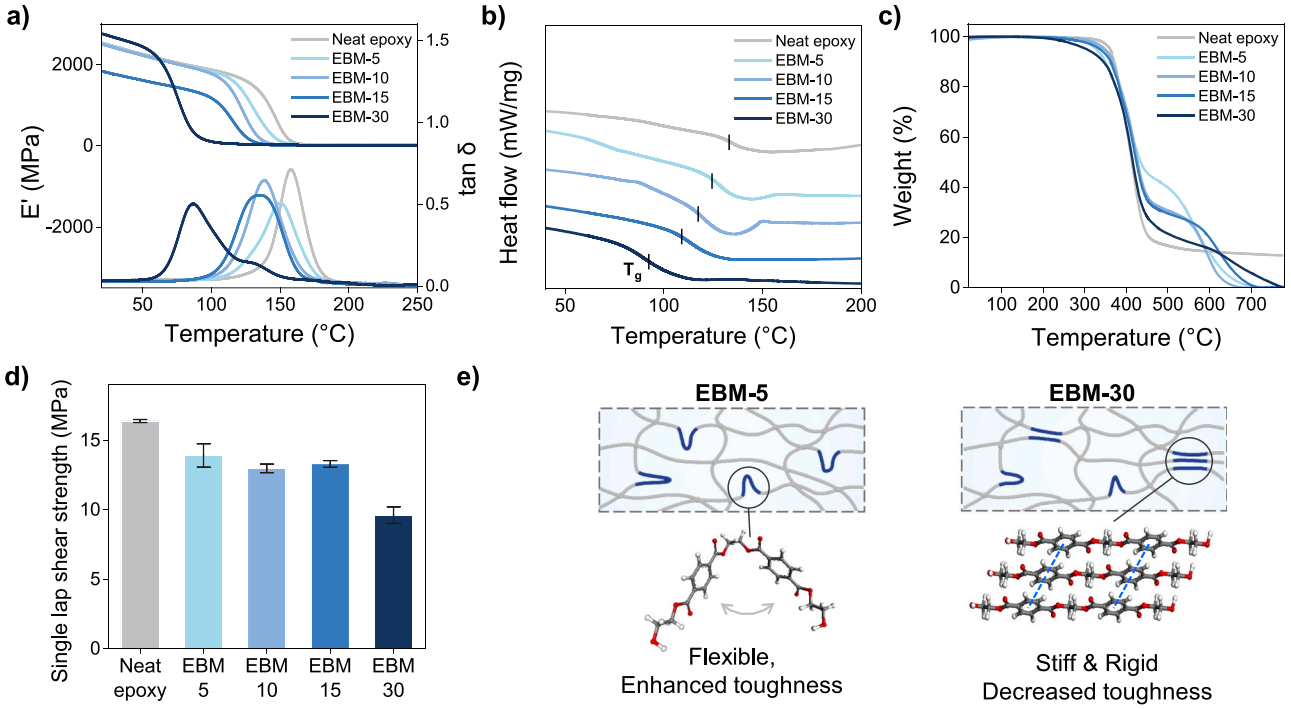


Fig. 6. (a) Storage modulus (E′) and tan δ as a function of temperature for EBM samples with varying mBHET content. (b) Glass transition temperatures (T_g) measured by DSC for epoxy formulations with different mBHET loadings. (c) Thermogravimetric analysis (TGA) curves of Neat epoxy and EBM samples. (d) Lap shear strength of Neat epoxy and EBM adhesives bonded to steel substrates. (e) Schematic illustration of the epoxy network in EBM-5 and EBM-30, presented as a conceptual model to describe the effect of mBHET content on cross-linking density and network morphology.

Table 2
Summary of thermal and mechanical properties, including cross-linking density, for EBM samples.

Sample name	E'_e (MPa)	$T_{g,DMA}$ (°C)	$T_{g,DSC}$ (°C)	ρ (x10^{-4}mol/cm^3)	T_{d5} (°C)	Lap shear strength (MPa)
Neat epoxy	25.17	158.1	134.3	20.96	358.7	16.40 ± 0.1
EBM-5	18.82	150.0	133.0	15.93	328.2	13.92 ± 0.8
EBM-10	12.6	138.5	122.5	10.93	344.8	12.99 ± 0.3
EBM-15	8.79	134.8	114.8	7.69	336.1	13.31 ± 0.2
EBM-30	28.22	86.9	94.7	27.58	307.1	9.62 ± 0.6

observed, primarily due to increased formulation viscosity caused by the powdered additive, which can limit wetting and interfacial contact during bonding (*Fig. S21*) [65]. Such an increase in viscosity may also affect processability in high-speed dispensing processes, where reduced flowability can hinder efficient substrate wetting and uniform adhesive spreading. Nevertheless, even at elevated loadings the measured shear strengths remained within the typical range for structural epoxy adhesives (~15–20 MPa) [66–68]. In practical applications, this limitation can be mitigated by incorporating reactive diluents to improve flowability and wetting behavior, which will be considered in future work [69]. Taken together, these results demonstrate that small amounts of PET-derived mBHET provide a rare combination of (i) highly efficient toughening at low loading, (ii) only modest T_g and T_{d5} penalties, and (iii) retention of practical adhesive strength—supporting mBHET as a drop-in reactive additive that improves the overall property balance of industrial one-component epoxy systems.

3. Conclusion

In this work, we present a practical and sustainable strategy for toughening one-component epoxy thermosets through the direct utilization of BHET derived from glycolyzed PET waste as a reactive additive. The crude monomer–dimer mixture, obtained without additional purification or chemical modification, was successfully incorporated into an industrially relevant DGEBA/DICY epoxy formulation while preserving the simplicity and storage stability of the one-component curing system.

At a remarkably low loading of 5 wt%, mBHET achieved substantial improvements in mechanical performance, increasing tensile toughness and impact strength by approximately 4400% and 91%, respectively, while maintaining glass transition temperature and thermal stability within practical ranges (T_{d5} > 300 °C). These results demonstrate that pronounced toughening can be achieved with minimal thermal penalty, effectively mitigating the conventional trade-off between toughness and thermal integrity commonly associated with epoxy toughening strategies.

Comprehensive structural and mechanistic analyses revealed that mBHET functions as a reactive, network-integrated additive rather than a phase-separating modifier. AFM and SAXS analyses confirmed the absence of phase separation, whereas XRD, XPS, and MD simulations revealed that the toughening efficiency is governed by stacking-dependent conformational behavior of the BHET species. At low loadings, folded and flexible dimer structures promote efficient energy dissipation and suppress crack propagation. At higher concentrations, increased π–π interactions between aromatic units induce rigidification and reduce toughness. Notably, the combined presence of BHET monomer and dimer provides a synergistic effect: monomer units disrupt excessive dimer stacking and preserve conformational flexibility, while dimers serve as effective molecular energy-dissipation elements. This intrinsic synergy enables superior performance of the crude mixture compared to either component alone.

Beyond its mechanical advantages, this work highlights a scalable and industrially relevant route for polymer upcycling. By directly employing unrefined PET glycolysis products as drop-in reactive additives, substantial performance enhancement can be achieved without

complex synthesis, purification, or reformulation of existing epoxy systems. The ability to improve fracture resistance while maintaining thermal stability and adhesive performance underscores the potential of mBHET as a commercially viable toughening agent for structural epoxy applications.

Overall, this study establishes that the inherent compositional heterogeneity of glycolysis-derived BHET species can be leveraged to construct a molecular architecture that promotes effective epoxy toughening. This approach not only advances the design of high-performance epoxy materials but also provides a sustainable pathway for transforming plastic waste into value-added functional additives. While this work investigated the synergistic toughening mechanism based on the experimentally obtained monomer-to-dimer ratio of approximately 1:0.4, the co-existence and complementary roles of these monomer and dimer species suggests that further tailoring of the monomer-to-dimer ratio could yield diverse and customizable materials properties. Future work directed toward controlling glycolysis conditions, and extending this strategy to bio-based epoxy systems may enable the development of fully sustainable, yet mechanically robust thermosetting materials.

CRediT authorship contribution statement

Gyuri Kim: Writing – original draft, Visualization, Validation, Investigation, Formal analysis. **Hongdeok Kim:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis. **Yungyeong Lee:** Writing – review & editing, Writing – original draft, Visualization, Supervision. **Joonmyung Choi:** Writing – review & editing, Supervision. **Min Sang Kwon:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2026.178661.

Data availability

Data will be made available on request.

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